**P0DAD-00、P0DCA-00、P0DE6-F0、P0DE6-F1、P0DE6-F2、P0DE6-F3、P0DE7-F0、P0DE7-F1、P0DE7-F2、P0DE7-F3、P1B02-00、P1B11-F0、C0111-96、C0112-96、P1B11-F1、C1001-04、P1E00-F0、 P1B44-00、 P1B48-00**

| **Fault diagnosis code** |
|---|
| P0DAD-00：balance circuit open |
| P0DCA-00：balance circuit short |
| P0DE6-F0：Single Cell undervoltage level1 |
| P0DE6-F1：single cell undervoltage level2 |
| P0DE6-F2：single cell undervoltage level3 |
| P0DE6-F3：single cell extremely undervoltage |
| P0DE7-F0：Single Cell Over Voltage Level 1 |
| P0DE7-F1：Single Cell Over Voltage Level 2 |
| P0DE7-F2：Single Cell Over Voltage Level 3 |
| P0DE7-F3：Single Cell Over Voltage Extremely |
| P1B02-00：BMS unintended reset |
| P1B11-F0：Cell voltage sample open wire fault |
| P1B11-F1：Cell voltage sensor fault |
| P1E00-F0：Cell unbalance |
| P1B44-00：balance circuit over temperature |
| P1B48-00：balance circuit temperature invalid fault |
| **Detection tools** |
| 1 Multimeter and diagnostic apparatus |
| **Possible causes** |
| • BMS fault<br>• BMS balance function fault<br>• Battery pack damage |
| **Conditions for fault code operation** |
| Turn the ignition switch to ON position. |
| **Conditions for fault code setting** |
| P0DAD-00 |
| BMS detects that the balance circuit open fault flag is valid&& CMC does not have a short circuit fault |
| P0DCA-00 |
| BMS detects that the balance circuit short fault flag is valid |
| P0DE6-F0 |

| |
|---|
| Unit voltage is lower than the alarm value:<br>Vmin ≤ 2800mV （Tmin>20 ℃）<br>Vmin ≤ 2500mV （10 ℃< Tmin ≤ =20 ℃）<br>Vmin ≤ 2300mV （0 ℃< Tmin ≤ 10 ℃）<br>Vmin ≤ 2100mV （-5 ℃< Tmin ≤ 0 ℃）<br>Vmin ≤ 2100mV （0 ℃< Tmin ≤ -5 ℃ )<br>Vmin ≤ 2100mV （Tmin ≤ -10 ℃ ) |
| P0DE6-F1 |
| Unit voltage is lower than the protection threshold:<br>Vmin ≤ 2500mV （Tmin>20 ℃）<br>Vmin ≤ 2350mV （10 ℃< Tmin ≤ =20 ℃）<br>Vmin ≤ 2100mV （0 ℃< Tmin ≤ 10 ℃）<br>Vmin ≤ 1900mV （-5 ℃< Tmin ≤ 0 ℃）<br>Vmin ≤ 1900mV （0 ℃< Tmin ≤ -5 ℃ )<br>Vmin ≤ 1900mV （Tmin ≤ -10 ℃ ) |
| P0DE6-F2 |
| Unit voltage is lower than the protection threshold1:<br>Vmin ≤ 2300mV （Tmin>20 ℃）<br>Vmin ≤ 2200mV （10 ℃< Tmin ≤ =20 ℃）<br>Vmin ≤ 1900mV （0 ℃< Tmin ≤ 10 ℃）<br>Vmin ≤ 1700mV （-5 ℃< Tmin ≤ 0 ℃）<br>Vmin ≤ 1700mV （0 ℃< Tmin ≤ -5 ℃ )<br>Vmin ≤ 1700mV （Tmin ≤ -10 ℃ ) |
| P0DE6-F3 |
| Unit voltage is lower than the protection threshold2:<br>Vmin ≤ 1800mV （Tmin>20 ℃）<br>Vmin ≤ 1800mV （10 ℃< Tmin ≤ =20 ℃）<br>Vmin ≤ 1500mV （0 ℃< Tmin ≤ 10 ℃）<br>Vmin ≤ 1500mV （-5 ℃< Tmin ≤ 0 ℃）<br>Vmin ≤ 1500mV （0 ℃< Tmin ≤ -5 ℃ )<br>Vmin ≤ 1500mV （Tmin ≤ -10 ℃ ) |
| P0DE7-F0 |
| Unit voltage is higher than the alarm value:<br>Vmax ≥ 3800mV |
| P0DE7-F1 |

| |
|---|
| Unit voltage is higher than the protection threshold:<br>Vmax ⩾ 3850mV |
| P0DE7-F2 |
| Unit voltage is higher than the safe threshold1:<br>Vmax ⩾ 3900mV |
| P0DE7-F3 |
| Unit voltage is higher than the safe threshold2:<br>Vmax ⩾ 4200mV |
| P1B02-00 |
| BMS detects Reset during system running: number of unexpected restarts of BMS ⩾ 1 |
| P1B11-F0 |
| The core voltage sampling line is short: (17823 sample line dropped diagnosis result is valid)&&(the voltage of either of the two adjacent cells of the dropped sampling line is less than or equal to 0.25V) |
| P1B11-F1 |
| The sampling voltage of the cell exceeds the normal voltage range of the cell: when no hardware fault occurs, the voltage of a unit exceeds the upper limit(over 0V~4.8V) |
| P1E00-F0 |
| The difference between the maximum SOC and minimum SOC exceeds the allowable range: (SOC_Max - SOC_Min) ⩾ 15% |
| P1B44-00 |
| Balance circuit temperature sampling is higher than protection threshold: balance circuit temperature >90 ℃ && balance circuit temperature valid |
| P1B48-00 |
| Balance circuit temperature sampling invalid: under the daisy chain upgraded: balance circuit temperature sampling invalid |
| **To set the effect of a fault code condition** |
| Cell function is limited |

| **Troubleshooting steps** | | |
|---|---|---|
| Step 1 | General inspection. | |
| | | 1 Check battery capacity.<br>2 Check the harness plug of BMS for damage, poor contact, aging and looseness.<br>3 Check the harness plug of battery pack high voltage for damage, poor contact, aging and looseness.<br>4 Check whether the result is normal or not? |
| | | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | | |
| Step 2 | Check fault codes. | |
| | | 1 Connect the fault diagnosis instrument to the diagnosis interface.<br>2 Turn the ignition switch to ON position.<br>3 Diagnose the BMS with diagnostic apparatus.<br>4 Is there any other fault code except for P0DAD-00、P0DCA-00、P0DE6-F0、P0DE6-F1、P0DE6-F2、P0DE6-F3、P0DE7-F0、P0DE7-F1、P0DE7-F2、P0DE7-F3、P1B02-00、P1B11-F0、P1B11-F1、P1E00-F0、C0062-08、P1B44-00、P1B48-00? |
| | | **YES →** Refer to:DTC Summary list(BMS). |
| **NO ↓** | | |
| Step 3 | Carry out automatic balance operation of battery pack cell. | |
| | | 1 According to the maintenance process of the manufacturer, the battery pack cells is automatically balanced.<br>2 Clear the BMS Fault code and re-use the diagnostic instrument to diagnose BMS.<br>3 Test whether the faulted code is existed. |
| | | **YES →** Diagnosis end. |
| **NO ↓** | | |

| | |
|---|---|
| Step 4 | Check the connecting cable of BMS. |
| | 1 Turn the ignition switch to OFF.<br>2 Make protective preparations according to the safety operation specifications of the high voltage voltage in the maintenance manual.<br>3 Disconnect the harness plug battery pack high voltage.<br>4 Disconnect the high voltage harness plug of BMS.<br>5 Check whether the BMS connection cable is rotten, damaged, and water enters.<br>6 Check whether the result is normal or not? |
| | **NO →** Repair or replace the faulted parts. |
| **YES ↓** | |
| Step 5 | Perform initialization. |
| | 1 Turn the ignition switch to OFF.<br>2 Connect the fault diagnosis instrument to the diagnosis interface.<br>3 Turn the ignition switch to ON.<br>4 The BMS module was initialized with the diagnostic instrument.<br>5 Can the BMS be initialized and is the BMS restored? |
| | **NO →** Repair or change the BMS module. |
| **YES ↓** | |
| Step 6 | Test whether BMS is norm. |
| | 1 Turn the ignition switch to OFF.<br>2 Disconnect and check the harness plug CH1000_1 of BMS.<br>3 Check and repair.<br>• Corrosion(install new adapting piece or terminal-cleaning module and pin).<br>• Pin damaged or bent-install new terminal/pin.<br>• Fly out of pins-install new pins as needed.<br>4 Reconnect the BMS plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed.<br>5 Run the system and determine if the problem persists? |
| | **YES →** Replace the BMS. |
| **NO ↓** | |
| At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems. | |

| **Next Step ↓** |
| --- |
| Diagnosis end. |